

[97]

Exam 4: NE533: Nuclear Fuel Performance

Show all work. Label question number in your response. Pay attention to units. Point values correspond to expected depth of response.

1. (18 pts) A ZIRLO cladding tube is in reactor at 625 K for 500 days. The initial wall thickness is 600 μm .
 - a) Estimate the oxide thickness after this time?
 - b) Assuming the hydrogen pickup fraction is 14%, what is the weight ppm of hydrogen in the cladding? Assume $\text{PBR} = 1.56$, $\rho_{\text{Zr}} = 6.5 \text{ g/cc}$, $\rho_{\text{ZrO}_2} = 5.68 \text{ g/cc}$.
2. (2 pts) What is the rate-limiting step in the aqueous corrosion of Zr cladding?
3. (8 pts) Where do hydrides form in the cladding? Why? What impacts do hydrides have?
4. (12 pts) What is a RIA? What is a typical RIA in a PWR or BWR. Describe what happens during a RIA.
5. (12 pts) What is a LOCA? Describe the impacts of a LOCA. How is it different than a RIA?
6. (4 pts) How does burnup impact the type of failure and the failure probability during an accident?
7. (6 pts) What are the four pathways to make the fuel/cladding system more accident tolerant?
8. (6 pts) List one near-term ATF concept and why it is of interest.
9. (6 pts) What happens to zirconium cladding when it is exposed to high temperature steam?
10. (6 pts) List three examples of limiting phenomena governing LWR operation.
11. (6 pts) What fuel performance and safety impacts does CRUD have?
12. (6 pts) List two water chemistry controls that have been implemented in LWRs, including why they were implemented.
13. (8 pts) List and describe two key microstructural phenomena that are unique to MOX fuels.

$$T_{\text{cat}} = 625 \text{ K}$$

$$t = 500 \text{ d}$$

$$\delta_0 = 600 \text{ nm}$$

$$t^*(d) = 6.62 \times 10^{-7} \exp\left(\frac{11.949}{t}\right)$$

18/18

$$t^*(d) = 133.0067287 \text{ d}$$

$$t \gg t^*(d)$$

$$\delta^*(nm) = 5.1 \exp(-554)$$

$$\delta^*(nm) = 2.11539285 \text{ nm}$$

$$K_L\left(\frac{1}{d}\right) = 7.48 \times 10^6 \exp(-13509)$$

$$K_L\left(\frac{1}{d}\right) = 0.0154174291 \frac{1}{\text{nm}}$$

$$\delta(nm) = \delta^* + K_L(t - t^*)$$

$$\boxed{\delta(nm) = 7.773486 \text{ nm}}$$

$$b. f = 14\%$$

$$PBR = 1.56$$

$$\rho_{Zr} = 6.5 \text{ g/cc}$$

$$\rho_{ZrO_2} = 5.68 \text{ g/cc}$$

$$f_{ZrO_2}^0 = \frac{2 \times M_o}{M_{Zr} + 2 \times M_o}$$

$$f_{ZrO_2}^0 = 0.2601626016$$

$$\frac{M_H}{M_o} = 0.0625$$

$$1 \text{ cm} = 1 \times 10^4 \text{ nm}$$

$$C_H^{\text{calc}}(\text{wt. ppm}) = \frac{2f \times \delta \times \rho_{\text{oxide}} \times f_{ZrO_2}^0 \times \frac{M_H}{M_o}}{(t - \frac{\delta}{PBR}) \times \rho_{\text{metal}}} \times 10^6$$

$$\boxed{C_H^{\text{calc}}(\text{wt. ppm}) = 62.47605 \text{ wt. ppm}}$$

~ 10 wt. ppm high
- not sure how

2. It is the diffusion or transport of oxygen through the oxide layer.

3. Hydrides form in the wall of the cladding. Hydrogen is formed from the oxidation reaction on the cladding surface and diffuses into the cladding. Hydrides form at areas of tensile stress and low temperature. Hydrides are less ductile than zirconium and can make the cladding more brittle.

4. RIA is a reactivity insertion accident. It is an accident caused by a sudden insertion of reactivity into the reactor. A typical RIA in a RWR is a control rod ejection accident (CREA) where a control rod is suddenly ejected out of the reactor. A typical RIA in a BWR is a control rod drop accident (CRDA) where a control rod suddenly drops out of the reactor. In a RIA, reactivity is suddenly inserted into the reactor usually due to a CREA or CRDA. This causes the fission rate and linear heat rate to rapidly increase. This causes the fuel temperature to rapidly increase. The increased heat flux causes departure from nucleate boiling (DNB) which causes film boiling and inefficient heat transfer. The decreased heat transfer causes the cladding temperature to increase to a point where the thermal stresses and temperature cause the cladding to burst. The fission products are then released. In extreme cases, the fuel can also melt and be released into the coolant.

5. LOCA is a loss of coolant accident. It causes there to be lower coolant flow in the reactor which causes the pressure to drop and the temperature to increase. The temperature increase causes the cladding

11/12

temperature to increase. This causes the cladding to swell and possibly burst. This blocks coolant flow which also increases the temperature. The damaged cladding can cause the release of fission products into the coolant. A LOCA is different from a RIA as a RIA occurs in only a few seconds whereas a LOCA occurs over several minutes.

6. Increased burnup ^{mostly} increases the magnitude of RIA and LOCA failures due to the increased fission products and cladding corrosion. It can cause more RIA failures. Increased burnup increases the probability of failure during RIA and LOCA accidents due to the increased fission products and cladding corrosion. - cladding embrittlement

7. 1. Improve steam corrosion kinetics -
2. Improve cladding properties -
3. Improve fuel properties -
4. Improve fission gas retention -

8. A near-term ATF concept is chromium coating for cladding. It is of interest as chromium and certain other metals are more resistant to corrosion than zirconium. ^{high T steam} This could make the cladding more resistant to corrosion and thus improve its heat transfer properties and make more efficient and possibly last for longer cycles. - sure

9. Zirconium cladding oxidizes when exposed to high temperatures. If its temperature increases to 863°C or higher, it begins to transform from the α -Zr to β -Zr crystal structure. It forms an α -Zr oxide layer and β -Zr inner layer. If the temperature increases to 1200°C, all of the zirconium is β -Zr and oxides and the cladding cracks.

10. 1. Power to melt -
2. cladding elongation and swelling -
3. Normal operation -

11. CRVD accumulates on the cladding surface and greatly decreases heat transfer from the cladding to the coolant. This increases the cladding and fuel temperatures. CRVD is also irradiated due to the Ni, Co, and other elements in it. CRVD can detach from the cladding surface and move to the heat exchanger. This can cause increased radioactive exposure to plant personnel.

12. LTH injection has been implemented in BWR's to lower the pH of the boric acid water which is used for reactivity control. The lower pH causes less corrosion to occur on the cladding which increases the heat transfer on the cladding. Hydrated water injection has been implemented in BWR's to limit intergranular structural chemical cracking (IGSCC). The decreased IGSCC causes more cladding strength and lower possible fission gas release.

13. The formation of the central void, columnar grains, equiaxed grains, and sintered grains is where the fuel forms four distinct regions radially due to a void region in center of the fuel and the movement of uranium and plutonium atoms down the thermal gradient. The increased fission product production from MAX fuel causes there to be more noble metals in the middle regions of the fuel and an increase of fuel swelling from fission products in the fuel.

- 2nd one is ... halfway

